

Supplementary Materials

Table S1. Country classification used for group-wise analysis.

Sr. No.	Country	Economy group	Classification basis	Analytical use
1	Australia	High-income economy	Economy-group classification used in this study	Group-wise descriptive comparison and scenario sensitivity
2	Austria			
3	Belgium			
4	Canada			
5	Czech Republic			
6	Denmark			
7	Estonia			
8	Finland			
9	France			
10	Germany			
11	Greece			
12	Hungary			
13	Iceland			
14	Ireland			
15	Israel			
16	Italy			
17	Japan			
18	Korea			
19	Latvia			
20	Lithuania			
21	Luxembourg			
22	Netherlands			
23	New Zealand			
24	Norway			
25	Portugal			
26	Slovak Republic			
27	Slovenia			
28	Spain			
29	Sweden			
30	Switzerland			

31	United Kingdom		Emerging economy					
32	United States							
33	Chile							
34	Colombia							
35	Costa Rica							
36	Mexico							
37	Türkiye							

Note: The analytical sample comprises 32 economies classified as high-income and 5 classified as emerging economies for the purposes of group-wise descriptive comparison and scenario sensitivity analysis. Economy-group classification is used as an analytical grouping variable and should not be interpreted as a direct measure of LTC-system maturity, institutional quality, financing sustainability, or policy performance. Because the emerging-economy subgroup contains only five economies, group-specific results are interpreted as exploratory.

Table S2. Missing-data summary and analytical treatment for the harmonized country-year panel.

Sr. No.	Variable	Analytical role	Expected country-year observations	Available observations	Missing observations	Missing (%)	Countries with available data	Observed year range	Recommended treatment
1	HealthExp_GDP / health expenditure burden	Primary expenditure outcome	518	518	0	0.00	37	2010–2023	Retain directly; complete coverage and no missing-value treatment required
2	Gov_Share	Supplementary public-financing indicator	518	41	477	92.08	3	2010–2023	Do not impute to the full panel; use only in exploratory restricted-sample financing analyses
3	Pop65	Population-aging indicator	518	518	0	0.00	37	2010–2023	Retain directly as a core predictor; no interpolation or imputation required
4	Dependency	Demographic-pressure indicator	518	518	0	0.00	37	2010–2023	Retain directly as a core predictor; no interpolation or imputation required
5	LTC_Workers	Supplementary LTC workforce-capacity indicator	518	58	460	88.80	16	2020–2023	Do not impute to the full panel; use only in exploratory restricted-sample workforce analyses
6	GDP_pc	Macroeconomic-capacity indicator	518	518	0	0.00	37	2010–2023	Retain directly as a core predictor; no interpolation or imputation required

7	Public_Exposure	Derived public-financing exposure indicator	518	41	477	92.08	3	2010–2023	Construct only where both HealthExp_GDP and Gov_Share are observed; treat as a supplementary restricted-sample descriptive indicator
8	FPI	Fiscal Pressure Index	518	518	0	0.00	37	2010–2023	Construct from HealthExp_GDP and Dependency; use as a derived descriptive or sensitivity indicator rather than an independent measure of fiscal sustainability
9	Worker_Ratio	Workforce adequacy ratio	518	58	460	88.80	16	2020–2023	Construct only where LTC_Workers and Pop65 are observed; use as a supplementary restricted-sample workforce indicator
10	Affordability	Macroeconomic affordability ratio	518	518	0	0.00	37	2010–2023	Construct from GDP_pc and HealthExp_GDP; use as a derived descriptive or sensitivity indicator

Note: The expected panel size comprises 37 economies observed annually from 2010 to 2023, yielding 518 country-year observations. In the final harmonized analytical panel, HealthExp_GDP, Pop65, Dependency, and GDP_pc have complete coverage; therefore, no interpolation or grouped imputation is required for the core econometric and machine-learning variables. Government financing share and LTC workforce capacity remain substantially incomplete and are not imputed to create full-panel observations. Analyses involving Gov_Share, LTC_Workers, Public_Exposure, or Worker_Ratio are restricted to country-years with observed parent variables and are interpreted as exploratory supplementary evidence. FPI and Affordability are deterministic transformations of completely observed core variables and are used for descriptive interpretation or sensitivity analysis rather than as independent measures of fiscal or LTC financing sustainability. Exact variable-specific missing-data treatment is further summarized in Supplementary Table S2C.

Table S2A. Source-variable clarification and interpretation of study variables

Manuscript variable	Clarified variable name	Source-variable meaning	Unit	Analytical role	Interpretation
HealthExp_GDP	Health expenditure burden	Total health expenditure as a share of gross domestic product	% of GDP	Primary outcome	Measures total health-system expenditure relative to national economic output. It is interpreted as health expenditure burden and not as direct LTC expenditure, LTC financing sustainability, or fiscal solvency.
Gov_Share	Government financing share	Public/government share of health financing	% or proportion	Supplementary restricted-sample financing indicator	Describes the public-sector financing share where observed. Because coverage is limited to 41 country-year observations from 3 countries, it is not used as a core full-panel

					predictor and does not support general cross-country financing conclusions.
Pop65	Population aged 65 years and above	Share or level of the population aged 65 years and above	Percentage or source-specific population measure	Core demographic predictor	Represents population aging and is used to examine its association with and predictive contribution to health expenditure burden. It should not be interpreted as a direct measure of LTC need.
Dependency	Old-age dependency ratio	Ratio of older persons relative to the working-age population	Ratio or percentage	Core demographic predictor	Represents demographic dependency pressure. It is used in the core econometric and predictive models and as a component of the derived Fiscal Pressure Index.
GDP_pc	Gross domestic product per capita	Economic output per person	Currency-based per-capita measure	Core macroeconomic predictor	Represents macroeconomic capacity and is used in expenditure-burden and income-elasticity analyses. It is also a component of the derived affordability ratio.
LTC_Workers	Long-term care workforce capacity	Formal LTC workforce availability reported in the source data	Count, rate, or source-specific workforce measure	Supplementary restricted-sample workforce indicator	Provides exploratory evidence on formal care-service capacity. Coverage is limited to 58 country-year observations from 16 countries, so it is not used as a core full-panel predictor.
Public_Exposure	Public financing exposure	Health expenditure burden combined with government/public financing share	Derived percentage/proportion	Supplementary derived financing indicator	Constructed only where Gov Share is observed. It provides an exploratory description of public financing exposure and is not interpreted as direct public LTC expenditure or an independent measure of fiscal sustainability.
FPI	Fiscal Pressure Index	Health expenditure burden combined with demographic dependency	Derived index	Derived descriptive/sensitivity indicator	Deterministic combination of health expenditure burden and dependency pressure. It is used for descriptive or sensitivity analysis and not as an independent measure of fiscal sustainability.
Worker_Ratio	Workforce adequacy ratio	LTC workforce relative to the older population	Derived ratio	Supplementary derived workforce indicator	Provides an exploratory measure of formal workforce availability relative to the older population. Its coverage is inherited from LTC_Workers and therefore remains restricted.

Affordability	Affordability ratio	GDP per capita relative to health expenditure burden	Derived ratio	Derived descriptive/sensitivity indicator	Provides a descriptive measure of macroeconomic capacity relative to health expenditure burden. Because it is deterministically constructed from existing variables, it is not an independent measure of affordability, solvency, or fiscal sustainability.
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Note: HealthExp_GDP, Pop65, Dependency, and GDP_pc constitute the complete-coverage core variables used in the principal econometric and predictive analyses. Gov_Share and LTC_Workers have substantially restricted country-year coverage and are therefore used only in supplementary exploratory analyses. Public_Exposure, FPI, Worker_Ratio, and Affordability are deterministic transformations of observed variables and are treated as descriptive or sensitivity indicators rather than independent measures of long-term care financing or fiscal sustainability.

Table S2B. Coverage profile and interpretation limits of LTC-specific financing and workforce indicators.

Variable	Available country-year observations	Unique countries with available data	Full panel reference	Coverage issue	Analytical treatment	Interpretation limit
Gov_Share	41	3	518 country-years from 37 countries	Very limited country coverage	Supplementary financing indicator; used cautiously in derived public-exposure analysis	Does not support broad group-level financing claims across all countries
LTC_Workers	58	16	518 country-years from 37 countries	Limited and uneven country-year coverage	Supplementary workforce-capacity indicator; used in restricted and sensitivity analysis	Does not support definitive cross-country workforce conclusions
Public_Exposure / public financing exposure	Depends on Gov_Share availability	Depends on Gov_Share availability	518 country-years from 37 countries	Inherits missingness from Gov_Share	Supplementary derived indicator	Interpreted as exploratory public financing exposure, not a full-panel estimate
Worker_Ratio	Depends on LTC_Workers and Pop65 availability	Depends on LTC_Workers' availability	518 country-years from 37 countries	Inherits missingness from LTC_Workers	Supplementary derived workforce indicator	Interpreted as exploratory service-capacity evidence

Table S2C. Variable-specific missing-data treatment in the final analytical panel.

Variable	Final available observations	Interpolated observations	Group-imputed observations	Missing retained	Analytical treatment
HealthExp_GDP	518	0	0	0	Primary outcome; complete panel
Pop65	518	0	0	0	Core predictor; complete panel
Dependency	518	0	0	0	Core predictor; complete panel

GDP_pc	518	0	0	0	Core predictor; complete panel
Gov_Share	41	0	0	477	Restricted-sample exploratory analysis only
LTC_Workers	58	0	0	460	Restricted-sample exploratory analysis only

Note: Missing government-financing and LTC-workforce observations were not filled to construct full-panel predictors. Analyses involving these variables use only country-years for which the corresponding observations are available. The core econometric and predictive models use variables with complete coverage across the 37-economy panel.

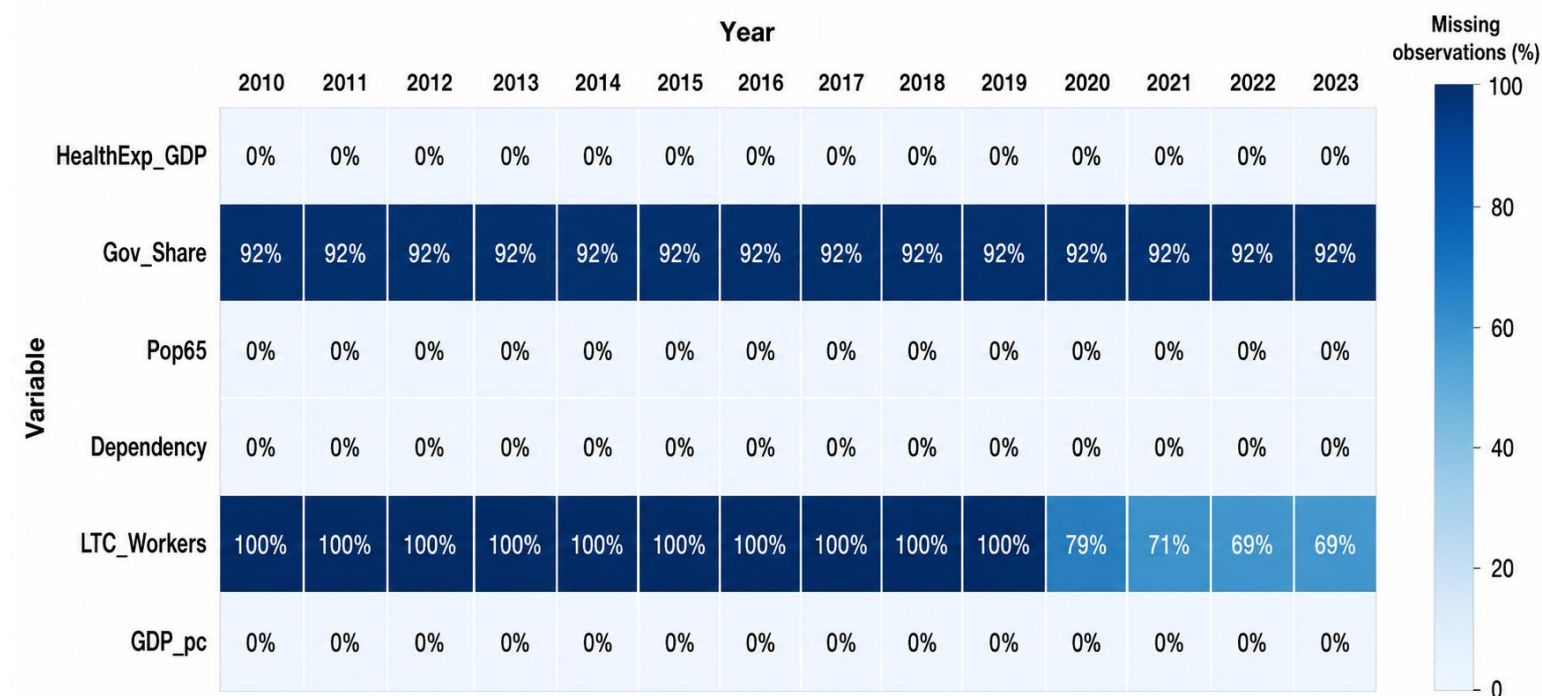


Figure S1. Missing-data pattern across variables and years.

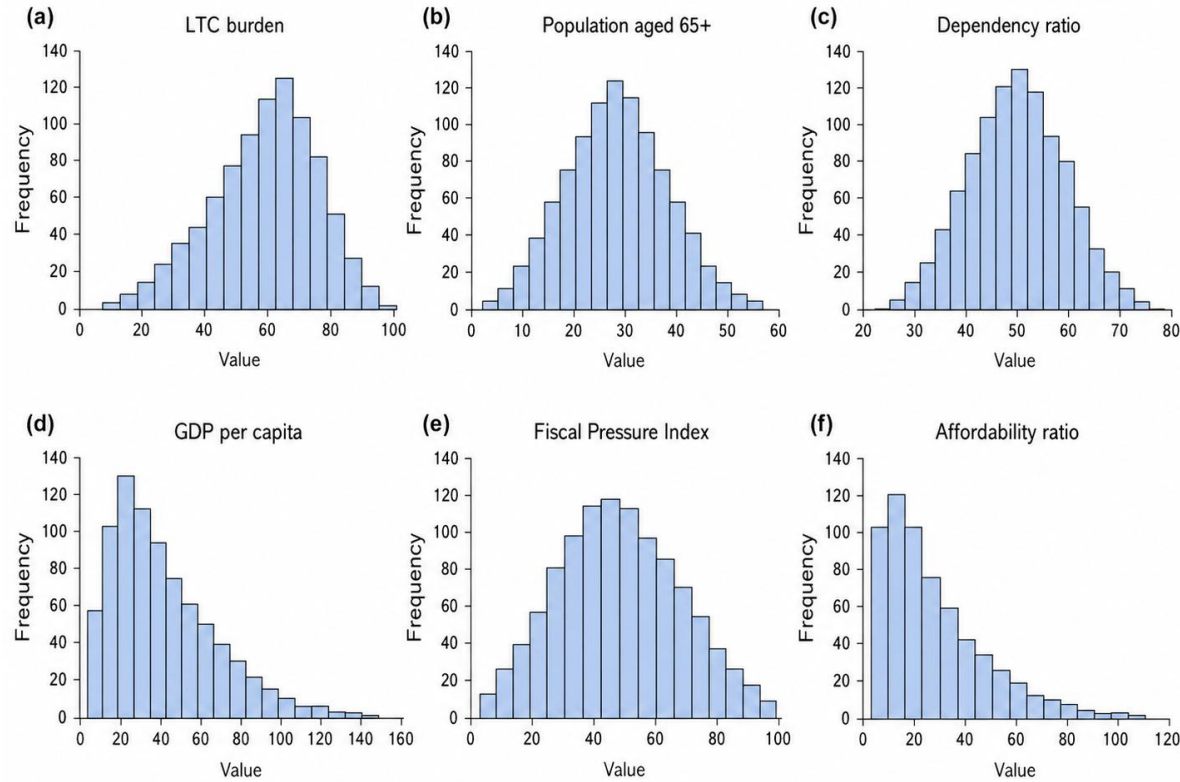
Figure S1 shows the annual missing-data pattern for the core and supplementary variables across the 2010–2023 analytical period. Health expenditure burden, population aged 65 years and above, old-age dependency ratio, and GDP per capita have complete coverage in the final 37-economy panel. In contrast, government financing share and LTC workforce capacity exhibit substantial and uneven missingness. Accordingly, these two variables are not imputed to construct full-panel predictors and are used only in exploratory restricted-sample analyses.

Table S3. Robustness strategy for core and restricted-sample analyses.

Robustness model	Sample specification	Missing-data treatment	Variables included	Purpose	Interpretation
R1	Core full analytical panel	No imputation required	HealthExp_GDP, Pop65, Dependency, GDP_pc, year/economy-group information as applicable	Benchmark econometric and predictive specification	Uses the complete-coverage 518-observation panel
R2	Core-variable specification	No imputation required	HealthExp_GDP, Pop65, Dependency, GDP_pc	Confirms results using only variables with complete coverage	Provides the principal full-panel demographic and macroeconomic evidence
R3	Financing-augmented restricted sample	Missing Gov_Share observations retained as missing; no full-panel imputation	Core variables + Gov_Share and related observed financing measures	Examines whether observed financing information provides additional exploratory evidence	Restricted to 41 observations from 3 countries; not generalized to the full panel
R4	Workforce-augmented restricted sample	Missing LTC_Workers observations retained as missing; no full-panel imputation	Core variables + LTC_Workers and Worker_Ratio where observed	Examines workforce-capacity information	Restricted to 58 observations from 16 countries; exploratory only
R5	High-income subsample	Core complete-coverage variables retained directly	Core variables	Examines whether full-panel relationships differ within the high-income group	Subgroup evidence; not a causal comparison
R6	Emerging-economy subsample	Core complete-coverage variables retained directly	Core variables	Examines whether relationships differ within the emerging-economy group	Exploratory because only five emerging economies are represented
R7	Alternative derived-outcome analysis	Parent-variable observations used directly	Public_Exposure, FPI, Affordability, or Worker_Ratio as applicable	Evaluates descriptive sensitivity to alternative constructed measures	Derived measures are not independent indicators of fiscal sustainability
R8	Scenario sensitivity checks	No imputation of financing or workforce variables	Pop65, Dependency, GDP_pc	Tests sensitivity to demographic/GDP trend multipliers and projection support	2024–2028 sensitivity analysis only; not a policy-effect or long-term forecast

The core specification therefore requires no imputation-based sensitivity comparison. Government-financing and LTC-workforce variables remain missing where unavailable and are used only in restricted-sample exploratory analyses.

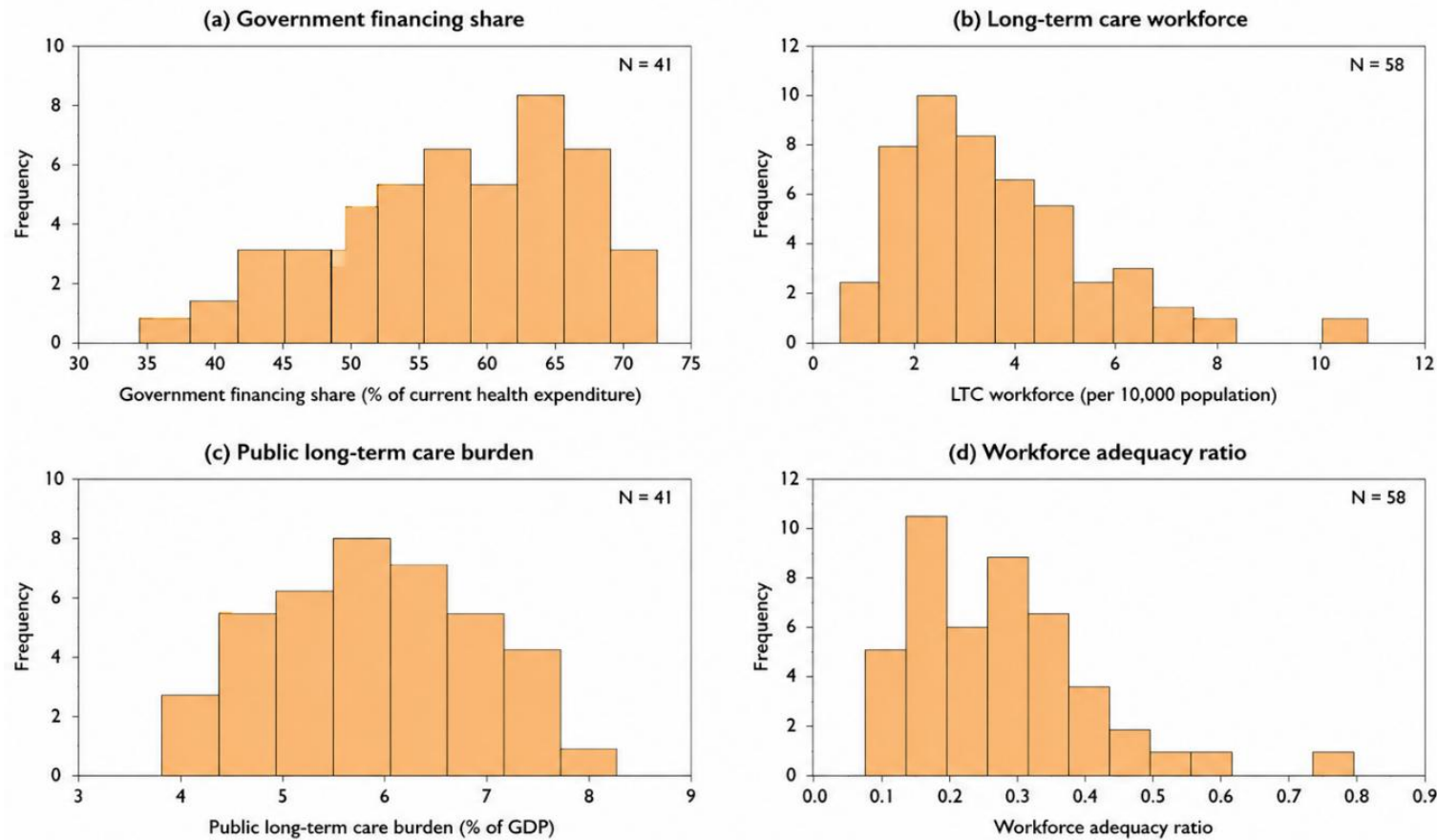
Distribution of Core Indicators



Supplementary Figure S2. Histograms of core demographic, fiscal, and macroeconomic indicators.

Figure S2 presents the distribution of key variables across a broad range of countries and years, including the health expenditure burden, the elderly population share, the dependency ratio, gross domestic product per capita, the Fiscal Pressure Index, and the affordability ratio. These histograms support the assessment of skewness, dispersion, and potential outliers prior to econometric and machine-learning modeling.

Distribution of financing and workforce indicators

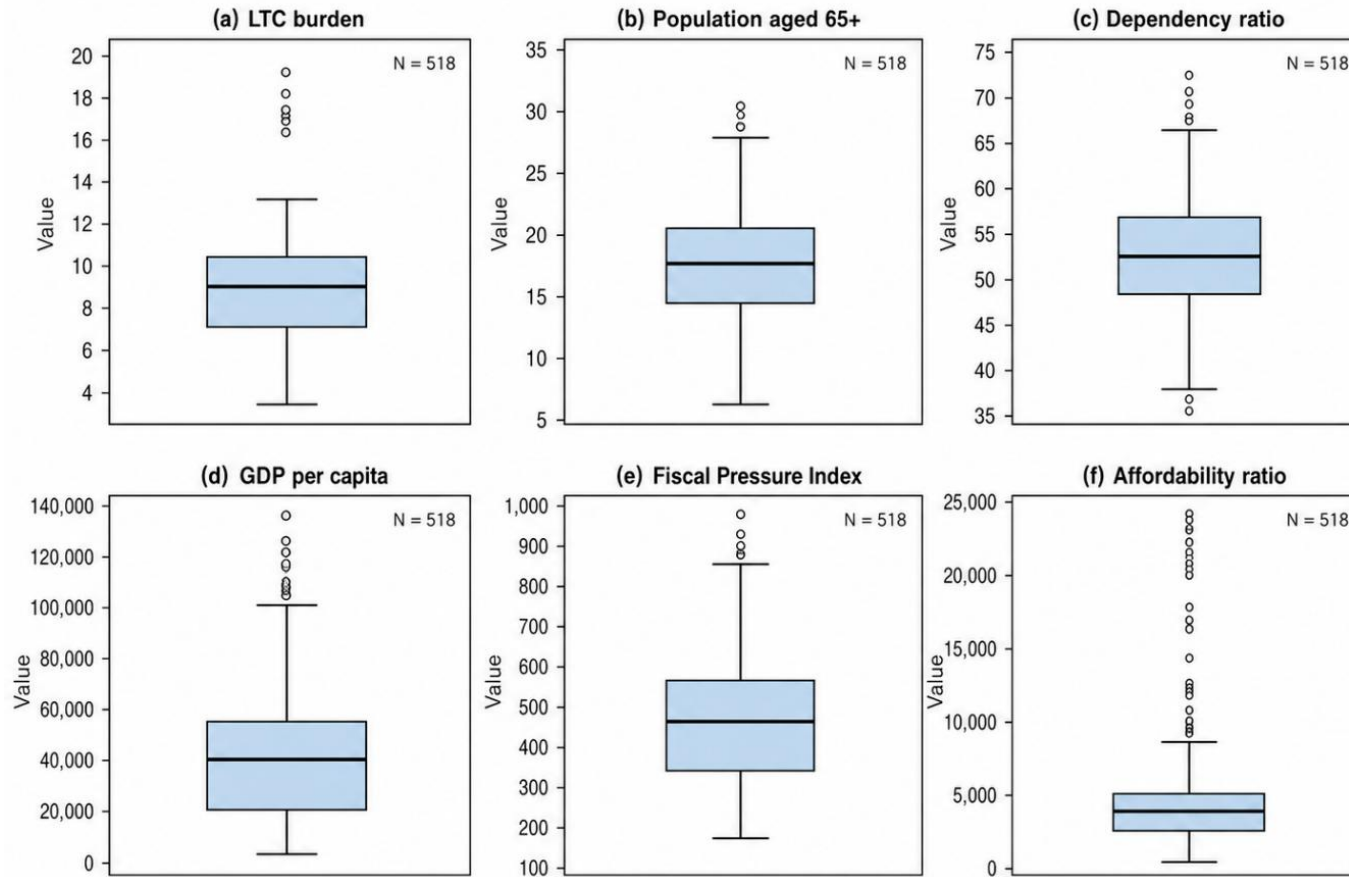


Note: N refers to the number of available country-year observations for each variable. Financing share and workforce-related variables have limited coverage and are interpreted with caution.

Figure S3. Histograms of financing and workforce-related indicators.

Figure S3 depicts the distribution of government financing share, long-term care workforce capacity, public financing exposure, and workforce adequacy ratio. These variables are shown separately because they have more limited country-year coverage and require cautious interpretation in the main analysis.

Distribution of core indicators

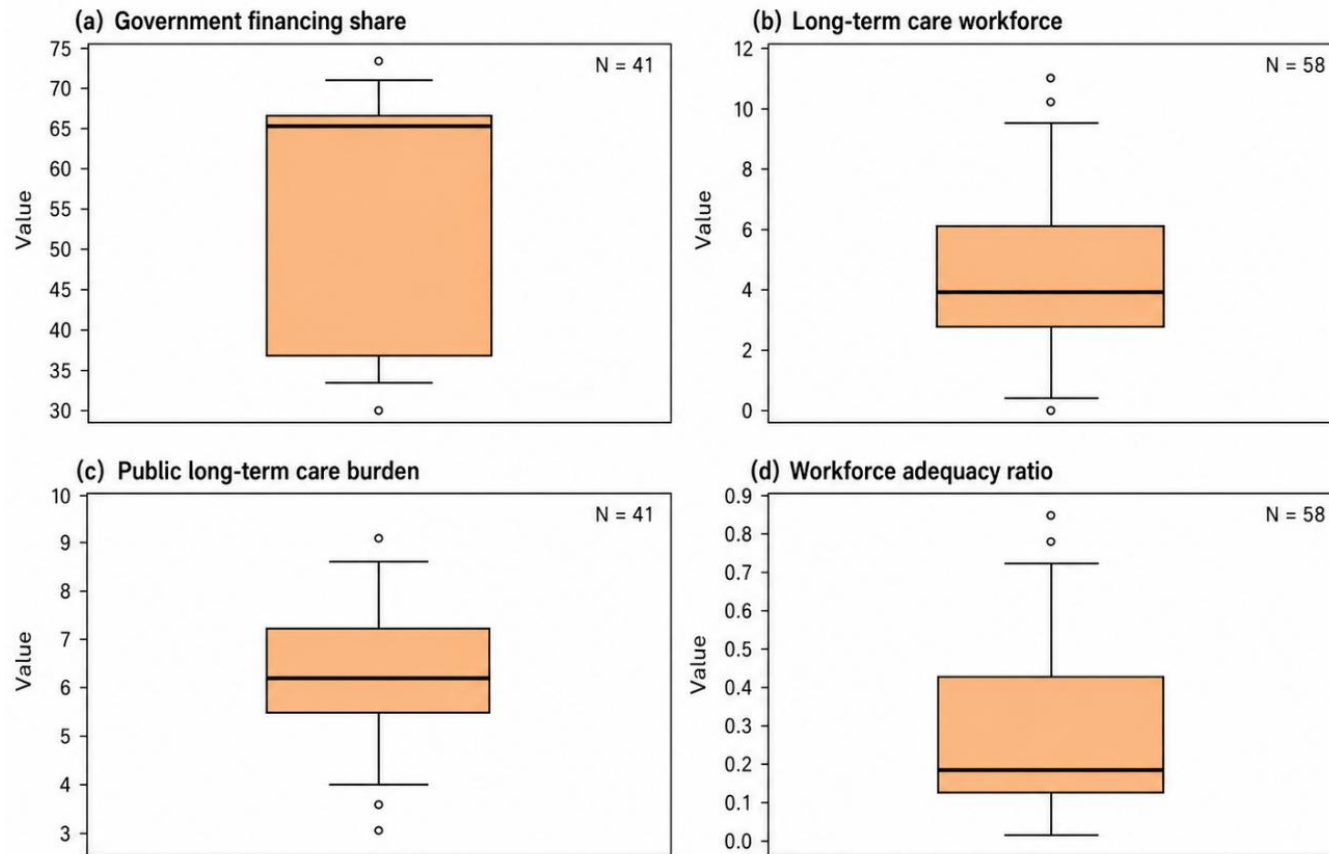


Note: Boxplots summarize median, interquartile range, whiskers, and outliers for each variable.

Figure S4. Boxplots of core demographic, fiscal, and macroeconomic indicators.

Figure S4 documents the distributional spread, cross-country dispersion, and potential outliers of the core indicators. Predictive model performance is evaluated separately using expanding-window validation.

Distribution of financing and workforce indicators



Note: Financing-share and workforce-related variables have more limited coverage and should be interpreted with caution.

Figure S5. Boxplots of financing and workforce-related indicators.

Figure S5 presents the boxplots for government financing share, long-term care workforce capacity, public financing exposure, and workforce adequacy ratio. These indicators show clear dispersion and outlier patterns, while their low observation counts support the cautious interpretation and the robustness checks applied in the modeling framework.

Table S4. Full correlation matrix of key study variables

Variable	Health expenditure burden	Gov. share	Pop. 65+	Dependency	GDP per capita	LTC workforce	Public financing exposure	FPI	Worker ratio	Affordability
Health expenditure burden	1.00	-0.87	0.42	0.31	0.33	-0.11	0.34	0.94	-0.23	-0.09
Gov. share	-0.87	1.00	-0.24	-0.59	-0.62	0.11	0.15	-0.87	0.11	-0.18
Pop. 65+	0.42	-0.24	1.00	0.56	0.16	-0.24	0.78	0.56	-0.50	-0.00
Dependency	0.31	-0.59	0.56	1.00	0.10	0.28	0.44	0.61	0.23	-0.06
GDP per capita	0.33	-0.62	0.16	0.10	1.00	0.40	0.58	0.31	0.42	0.88
LTC workforce	-0.11	0.11	-0.24	0.28	0.40	1.00	0.93	0.00	0.93	0.32
Public financing exposure	0.34	0.15	0.78	0.44	0.58	0.93	1.00	0.34	0.93	0.75
FPI	0.94	-0.87	0.56	0.61	0.31	0.00	0.34	1.00	-0.13	-0.08
Worker ratio	-0.23	0.11	-0.50	0.23	0.42	0.93	0.93	-0.13	1.00	0.40
Affordability	-0.09	-0.18	-0.00	-0.06	0.88	0.32	0.75	-0.08	0.40	1.00

Note: The Fiscal Pressure Index is mechanically related to the health expenditure burden because it is constructed from the health expenditure burden and the dependency ratio. Therefore, its high correlation with the primary outcome is expected and should not be interpreted as an independent empirical association. FPI is used as a derived fiscal-pressure indicator and as an alternative robustness outcome, not as an independent predictor of the health expenditure burden in the core model.

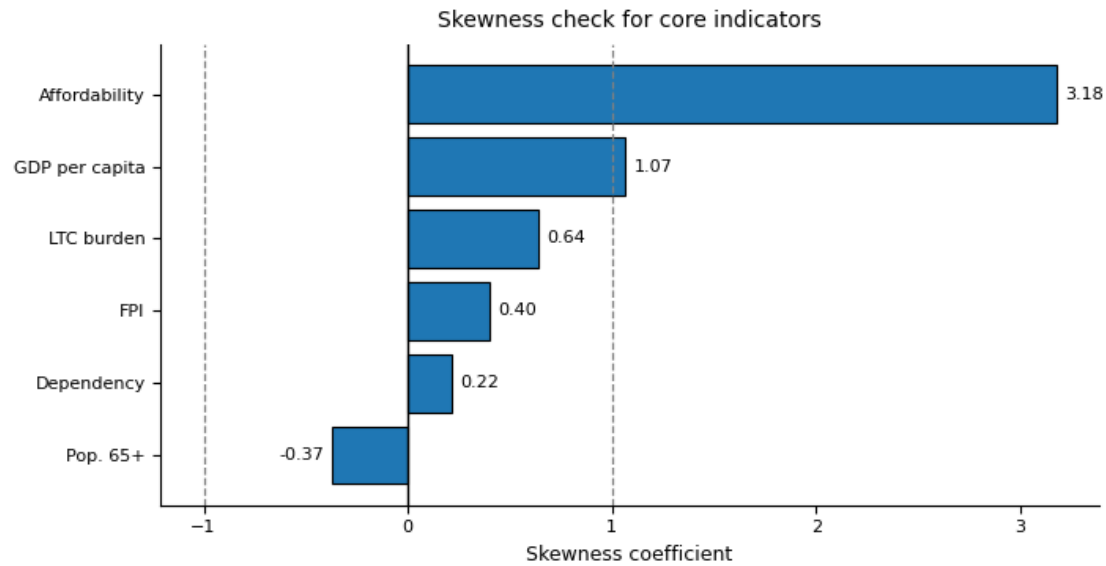


Figure S6. Skewness check for core demographic, fiscal, and macroeconomic indicators.

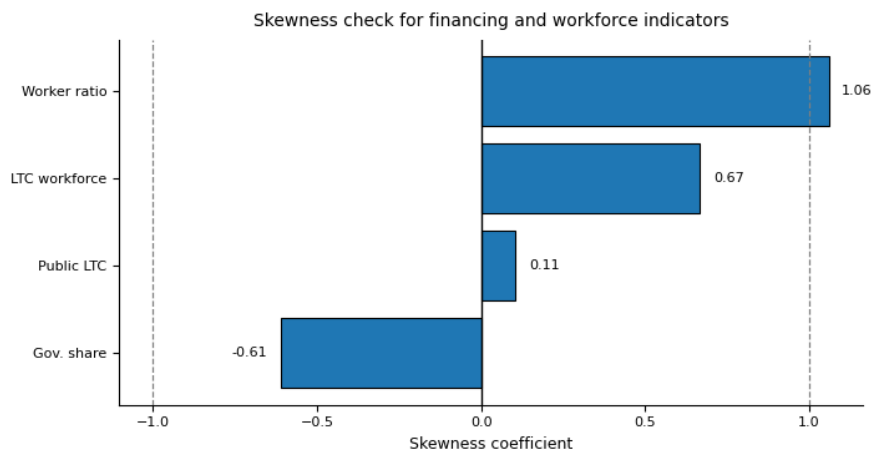


Figure S7. Skewness check for financing and workforce-related indicators.

Table S5. Full pooled log-log model results with country-clustered standard errors

Variable / model component	Coefficient	Clustered SE	t statistic	p-value	95% CI
Population aged 65+, elasticity	0.254	0.127	2.006	0.052	−0.003 to 0.511
Dependency ratio, elasticity	0.287	0.306	0.939	0.354	−0.333 to 0.908
GDP per capita, elasticity of expenditure burden	0.134	0.081	1.659	0.106	−0.030 to 0.297
Year effects	Included	—	—	—	—
Country effects	Not included	—	—	—	—
Country-clustered inference	Yes	—	—	—	—
Observations	518	—	—	—	—
Countries	37	—	—	—	—

Note: The dependent variable is the natural logarithm of health expenditure burden, measured as total health expenditure as a percentage of GDP. Continuous predictors are entered in natural logarithms; therefore, the reported coefficients are elasticities of the expenditure-to-GDP burden. Standard errors are clustered by country. Statistical inference uses 37 country clusters, with p-values and confidence intervals based on the corresponding cluster-level degrees of freedom. The aging coefficient is positive but is not statistically significant at the 5% level.

Table S6. Full fixed-effects log-log model results with country-clustered standard errors

Variable / model component	Coefficient	Clustered SE	t statistic	p-value	95% CI
Population aged 65+, elasticity	0.030	0.323	0.094	0.926	−0.624 to 0.685
Dependency ratio, elasticity	0.197	0.214	0.917	0.365	−0.238 to 0.632
GDP per capita, elasticity of expenditure burden	−0.143	0.145	−0.988	0.330	−0.438 to 0.151
Country fixed effects	Included	—	—	—	—
Year effects	Included	—	—	—	—
Country-clustered inference	Yes	—	—	—	—
Observations	518	—	—	—	—
Countries	37	—	—	—	—

Note: The dependent variable and continuous predictors are expressed in natural logarithms. The fixed-effects specification controls for time-invariant country characteristics and common year-specific effects. Standard errors are clustered by country. The coefficients therefore characterize conditional within-country relationships after accounting for country and year effects. None of the three core elasticities is statistically distinguishable from zero in this specification.

Table S7. Full random-effects log-log model results with country-clustered standard errors

Variable / model component	Coefficient	Clustered SE	t statistic	p-value	95% CI
Population aged 65+, elasticity	0.164	0.213	0.772	0.445	−0.267 to 0.595
Dependency ratio, elasticity	0.208	0.207	1.005	0.322	−0.211 to 0.627
GDP per capita, elasticity of expenditure burden	−0.088	0.122	−0.719	0.477	−0.335 to 0.160

Country random intercept	Included	—	—	—	—
Year effects	Included	—	—	—	—
Country-clustered inference	Yes	—	—	—	—
Observations	518	—	—	—	—
Countries	37	—	—	—	—

Note: The random-effects specification includes a country-level random intercept and year effects. The dependent variable and continuous predictors are entered in natural logarithms. Country-clustered standard errors are reported so that inference is robust to heteroskedasticity and within-country dependence. The random-effects estimates are reported for comparison with the pooled and fixed-effects specifications.

Table S7A. Within- and between-country elasticity estimates for health expenditure burden and per-capita health expenditure

Component	Population-aging elasticity of health expenditure burden	Dependency elasticity of health expenditure burden	Income elasticity of per-capita health expenditure
Between-country	0.247	0.306	1.141
Within-country	0.030	0.197	0.857

Note: Population-aging and dependency coefficients refer to elasticities of health expenditure burden, measured as health expenditure as a percentage of GDP. The income elasticity refers to the income elasticity of per-capita health expenditure with respect to GDP per capita. For income, the estimated 95% confidence intervals are 0.974–1.307 for the between-country elasticity and 0.572–1.141 for the within-country elasticity. Because both intervals include unity, the results do not support a definitive luxury-versus-necessity classification.

Table S8. Hausman specification test for the log-log panel models

Test component	Result
Null hypothesis	Random-effects estimator is consistent
Alternative hypothesis	Fixed-effects estimator is preferred
Common coefficients tested	ln(Pop65), ln(Dependency), ln(GDP pc)
Chi-square statistic	21.991
Degrees of freedom	3
p-value	<0.001
Decision	Reject the null hypothesis
Specification implication	Fixed effects are preferred for interpretation of within-country relationships

Note: The Hausman test compares the common slope coefficients from the fixed- and random-effects estimators. The conventional Hausman covariance framework is used for specification testing, while Tables S5–S7 report country-clustered standard errors for substantive statistical inference. This significant result supports emphasizing the fixed-effects estimates when interpreting within-country relationships.

Table S9. Heteroskedasticity diagnostic for the pooled log-log model

Test	Statistic	p-value	Decision	Interpretation
Breusch–Pagan LM test	83.543	<0.001	Reject homoskedasticity	Evidence of non-constant residual variance remains
Breusch–Pagan F test	6.021	<0.001	Reject homoskedasticity	Heteroskedasticity remains relevant after log transformation

Note: The Breusch–Pagan test was applied to the pooled log-log specification including the logged core predictors and year effects. The results indicate non-constant residual variance. Accordingly, Tables S5–S7 use country-level clustered standard errors. Log transformation is used for functional-form and elasticity interpretation rather than as a remedy for all residual heteroskedasticity.

Table S10. Panel serial-correlation diagnostic for the log specification

Diagnostic component	Result
Diagnostic approach	Wooldridge-type test for serial correlation in panel data
Number of countries	37
Usable differenced/lagged observations	444
Auxiliary residual-lag coefficient	0.038
Clustered SE of residual-lag coefficient	0.047
F statistic	132.115
p-value	<0.001
Decision	Reject the null of no first-order serial correlation
Inferential response	Use country-clustered standard errors

Note: The panel serial-correlation diagnostic indicates substantial within-country dependence in the error structure. Accordingly, pooled, fixed-effects, and random-effects coefficient inference uses country-clustered standard errors to account for within-country dependence. Under this Wooldridge-type auxiliary regression, the null hypothesis of no first-order serial correlation corresponds to testing the residual-lag coefficient against -0.5 rather than zero; the reported F statistic refers to this restriction.

Table S11. Variance inflation factor diagnostics for the logged core predictors

Predictor	VIF	Multicollinearity interpretation
ln(Population aged 65+)	1.523	Low multicollinearity
ln(Dependency ratio)	1.301	Low multicollinearity
ln(GDP per capita)	1.257	Low multicollinearity

Note: VIF values for the three logged continuous predictors are below commonly used diagnostic thresholds, indicating low multicollinearity among the core predictors. Derived indicators such as the Fiscal Pressure Index, public financing exposure, workforce adequacy ratio, and affordability ratio are not entered simultaneously with their parent variables in the core specification because they are deterministic transformations.

Table S12. Full expanding-window validation performance of candidate predictive models

Rank	Model	Mean validation RMSE	Mean MAE	Mean sMAPE (%)	Mean R ²	RMSE SD
1	CatBoost Regressor	1.238	0.765	8.155	0.723	0.254
2	LightGBM Regressor	1.268	0.788	8.465	0.706	0.300
3	XGBoost Regressor	1.425	0.873	9.465	0.634	0.240
4	Gradient Boosting Regressor	1.447	0.890	9.670	0.621	0.297
5	Random Forest Regressor	1.477	0.933	10.086	0.609	0.152
6	Log-linear regression	1.971	1.381	15.397	0.305	0.215
7	Elastic Net	2.109	1.552	17.337	0.205	0.157
8	Linear Regression	2.118	1.557	17.424	0.198	0.188

Note: Values are averages across five expanding-window validation folds with validation years 2017, 2018, 2019, 2020, and 2021. For every fold, models were trained only on observations preceding the validation year. Model-family and hyperparameter selection were based exclusively on development-period validation performance. The 2022–2023 observations were excluded from all values reported in this table and were reserved for the independent final test evaluation. CatBoost achieved the lowest mean validation RMSE and was therefore selected for final evaluation, SHAP interpretation, and conformal uncertainty assessment.

Table S13. Expanding-window validation performance of the validation-selected CatBoost model.

Validation fold	Training period	Training observations	Validation year	Validation observations	CatBoost RMSE
1	2010–2016	259	2017	37	1.199
2	2010–2017	296	2018	37	1.001
3	2010–2018	333	2019	37	1.067
4	2010–2019	370	2020	37	1.651
5	2010–2020	407	2021	37	1.270
Mean	—	—	—	—	1.238
Standard deviation	—	—	—	—	0.254

Note: The 2010–2021 observations constituted the model-development period, while observations from 2022–2023 were reserved as an untouched final test period. Expanding-window validation was conducted sequentially using 2017–2021 as validation years. For each fold, the model was trained exclusively on observations preceding the corresponding validation year, thereby preventing temporal leakage. Model-family and hyperparameter selection were based on mean validation RMSE across these five folds. CatBoost achieved a mean validation RMSE of 1.238 and was therefore retained as the final predictive model. The 2022–2023 test observations were not used for model or hyperparameter selection.

The final CatBoost configuration selected through development-period validation was: iterations = 500, learning_rate = 0.10, depth = 8, and l2_leaf_reg = 5.

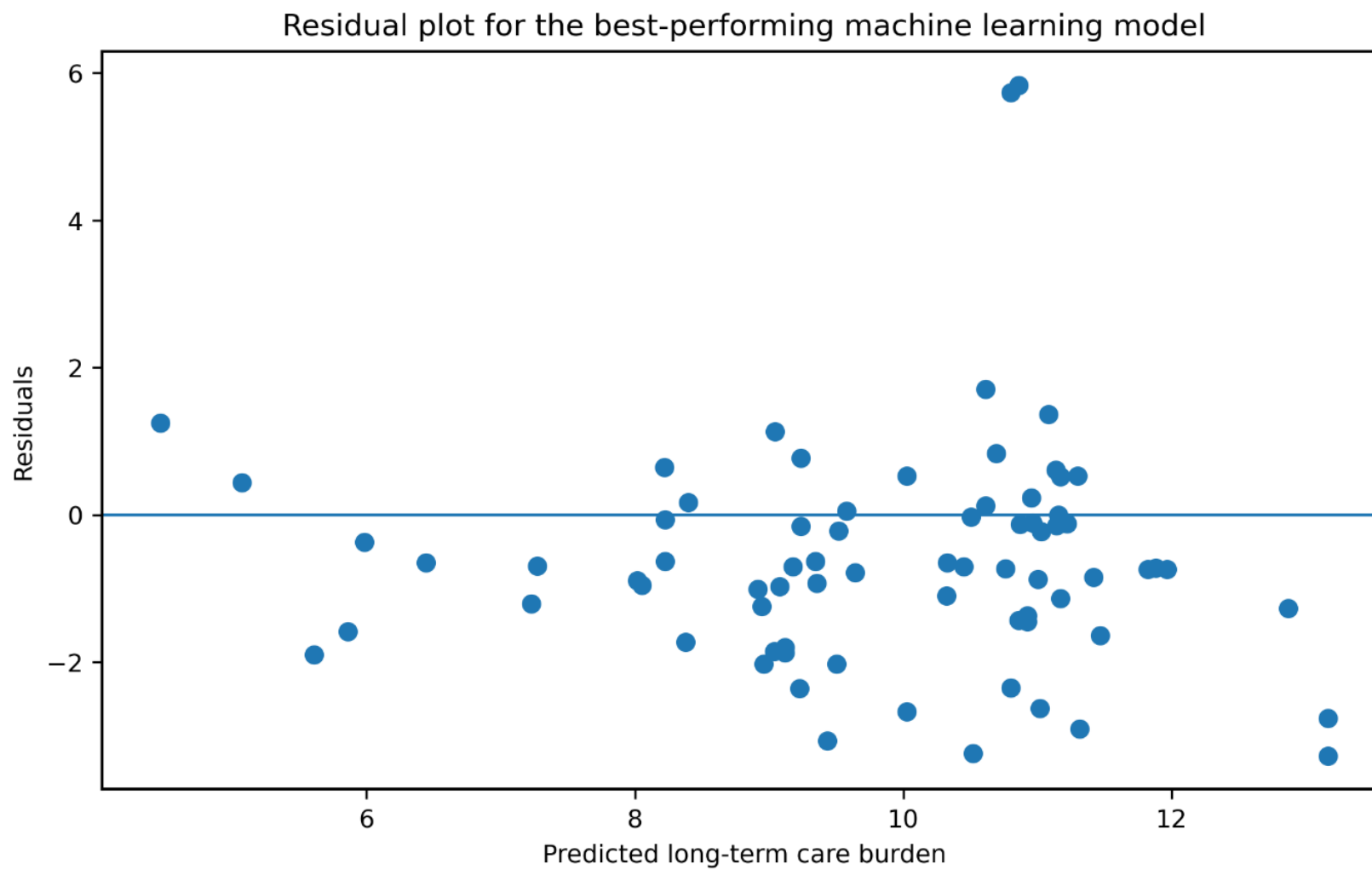


Figure S8. Residual plot for the best-performing machine learning model.

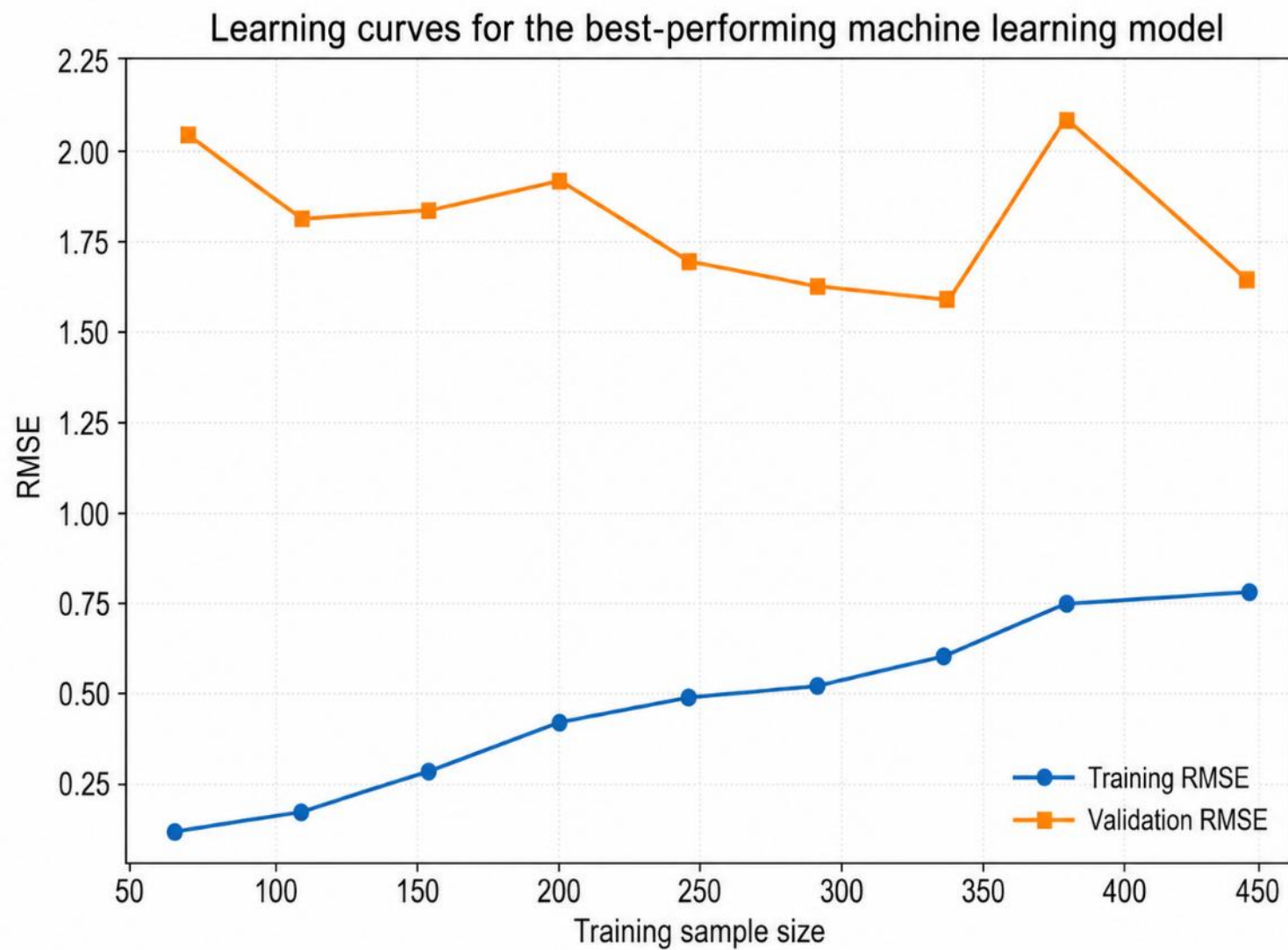


Figure S9. Learning curves for the best-performing machine learning model.

Table S14. Documented hyperparameter search-space design for development-period model tuning

Model	Hyperparameter	Search values/range	Purpose
Elastic Net	alpha	0.001, 0.01, 0.10, 1.0, 10.0	Controls overall regularization strength
Elastic Net	l1 ratio	0.10, 0.20, 0.50, 0.80, 1.00	Controls the balance between L1 and L2 penalties
Random Forest Regressor	n_estimators	100, 200, 300, 500	Number of trees
Random Forest Regressor	max_depth	3, 5, 7, 10, None	Controls maximum tree depth
Random Forest Regressor	min_samples_split	2, 5, 10	Minimum samples required for an internal-node split
Random Forest Regressor	min_samples_leaf	1, 2, 4	Minimum samples required in a terminal leaf
Random Forest Regressor	max_features	sqrt, log2, None	Number of features considered at each split
XGBoost Regressor	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling boosting step size
XGBoost Regressor	max_depth	3, 5, 7	Maximum tree depth
XGBoost Regressor	n_estimators	100, 200, 300	Number of boosting rounds
XGBoost Regressor	subsample	0.70, 0.85, 1.00	Row-subsampling proportion
XGBoost Regressor	reg_lambda	0, 1, 5	L2 regularization strength
XGBoost Regressor	min_child_weight	1, 3, 5	Minimum child-node weight
LightGBM Regressor	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling boosting step size
LightGBM Regressor	max_depth	3, 5, 7, -1	Tree-depth control
LightGBM Regressor	n_estimators	100, 200, 300	Number of boosting iterations
LightGBM Regressor	subsample	0.70, 0.85, 1.00	Row-subsampling proportion
LightGBM Regressor	reg_lambda	0, 1, 5	L2 regularization strength
LightGBM Regressor	min_child_samples	5, 10, 20	Minimum observations required in a leaf
CatBoost Regressor	iterations	100, 200, 300, 500	Number of boosting iterations
CatBoost Regressor	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling boosting step size
CatBoost Regressor	depth	4, 6, 8	Tree depth
CatBoost Regressor	l2_leaf_reg	1, 3, 5, 7	L2 leaf-regularization coefficient

Note: Hyperparameter evaluation was restricted to the 2010–2021 development period, with model assessment based on expanding-window validation. The 2022–2023 observations were reserved for independent final evaluation and were not used for model-family or hyperparameter selection. The final validation-selected CatBoost configuration was iterations = 500, learning_rate = 0.10, depth = 8, and l2_leaf_reg = 5. Gradient Boosting Regressor was included in the candidate-model comparison reported in Supplementary Table S12, but a separate Gradient Boosting hyperparameter-search space is not documented in the available tuning records and is therefore not presented in this table. RMSE = root mean squared error.

Table S15. Prespecified Grid Search design under expanding-window validation

Model	Hyperparameters included	Candidate values	Total candidate combinations	Evaluation criterion	Final-test involvement
Elastic Net	alpha; l1_ratio	alpha: 0.001, 0.01, 0.10, 1.0, 10.0; l1_ratio: 0.10, 0.20, 0.50, 0.80, 1.00	25	Lowest mean expanding-window validation RMSE	None
Random Forest Regressor	n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features	n_estimators: 100, 200, 300, 500; max_depth: 3, 5, 7, 10, None; min_samples_split: 2, 5, 10; min_samples_leaf: 1, 2, 4; max_features: sqrt, log2, None	540	Lowest mean expanding-window validation RMSE	None
XGBoost Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight	learning_rate: 0.01, 0.05, 0.10; max_depth: 3, 5, 7; n_estimators: 100, 200, 300; subsample: 0.70, 0.85, 1.00; reg_lambda: 0, 1, 5; min_child_weight: 1, 3, 5	729	Lowest mean expanding-window validation RMSE	None
LightGBM Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples	learning_rate: 0.01, 0.05, 0.10; max_depth: 3, 5, 7, -1; n_estimators: 100, 200, 300; subsample: 0.70, 0.85, 1.00; reg_lambda: 0, 1, 5; min_child_samples: 5, 10, 20	972	Lowest mean expanding-window validation RMSE	None
CatBoost Regressor	iterations; learning_rate; depth; l2_leaf_reg	iterations: 100, 200, 300, 500; learning_rate: 0.01, 0.05, 0.10; depth: 4, 6, 8; l2_leaf_reg: 1, 3, 5, 7	144	Lowest mean expanding-window validation RMSE	None

Note: This table documents the prespecified Grid Search parameter design used for development-period model evaluation. Expanding-window validation used 2017, 2018, 2019, 2020, and 2021 as sequential validation years, with each model trained only on observations preceding the corresponding validation year. Candidate configurations were assessed using mean validation RMSE. The 2022–2023 final test period was excluded from parameter selection and reserved for independent evaluation. Candidate-model validation performance is reported in Supplementary Table S12, fold-wise CatBoost validation results in Supplementary Table S13, and the independent final CatBoost evaluation in Supplementary Table S23.

Table S16. Prespecified Random Search framework under expanding-window validation

Model	Parameters sampled	Sampling domain	Planned candidate budget	Evaluation criterion	Analytical role
Elastic Net	alpha; l1_ratio	Parameter domain defined in Table S14	Up to 25 configurations	Lowest mean expanding-window validation RMSE	Regularized linear tuning benchmark
Random Forest Regressor	n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features	Parameter domain defined in Table S14	Up to 30 randomly sampled configurations	Lowest mean expanding-window validation RMSE	Efficient exploration of nonlinear tree configurations

XGBoost Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight	Parameter domain defined in Table S14	Up to 30 randomly sampled configurations	Lowest mean expanding-window validation RMSE	Efficient exploration of boosting configurations
LightGBM Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples	Parameter domain defined in Table S14	Up to 30 randomly sampled configurations	Lowest mean expanding-window validation RMSE	Efficient exploration of leaf-wise boosting configurations
CatBoost Regressor	iterations; learning_rate; depth; l2_leaf_reg	Parameter domain defined in Table S14	Up to 30 randomly sampled configurations	Lowest mean expanding-window validation RMSE	Alternative exploration of CatBoost complexity and regularization

Note: This table documents the Random Search framework specified for development-period tuning. Candidate configurations were evaluated using the same expanding-window validation structure used for model selection, with validation years 2017–2021 and training restricted to earlier observations. The 2022–2023 final test period was excluded from all tuning decisions. Because the available tuning documentation does not identify a reproducible executed random-state value or complete candidate-by-candidate Random Search log, the table reports the prespecified sampling framework and candidate budget rather than claiming an exact trial history.

Table S17. Prespecified Optuna Tree-structured Parzen Estimator optimization framework

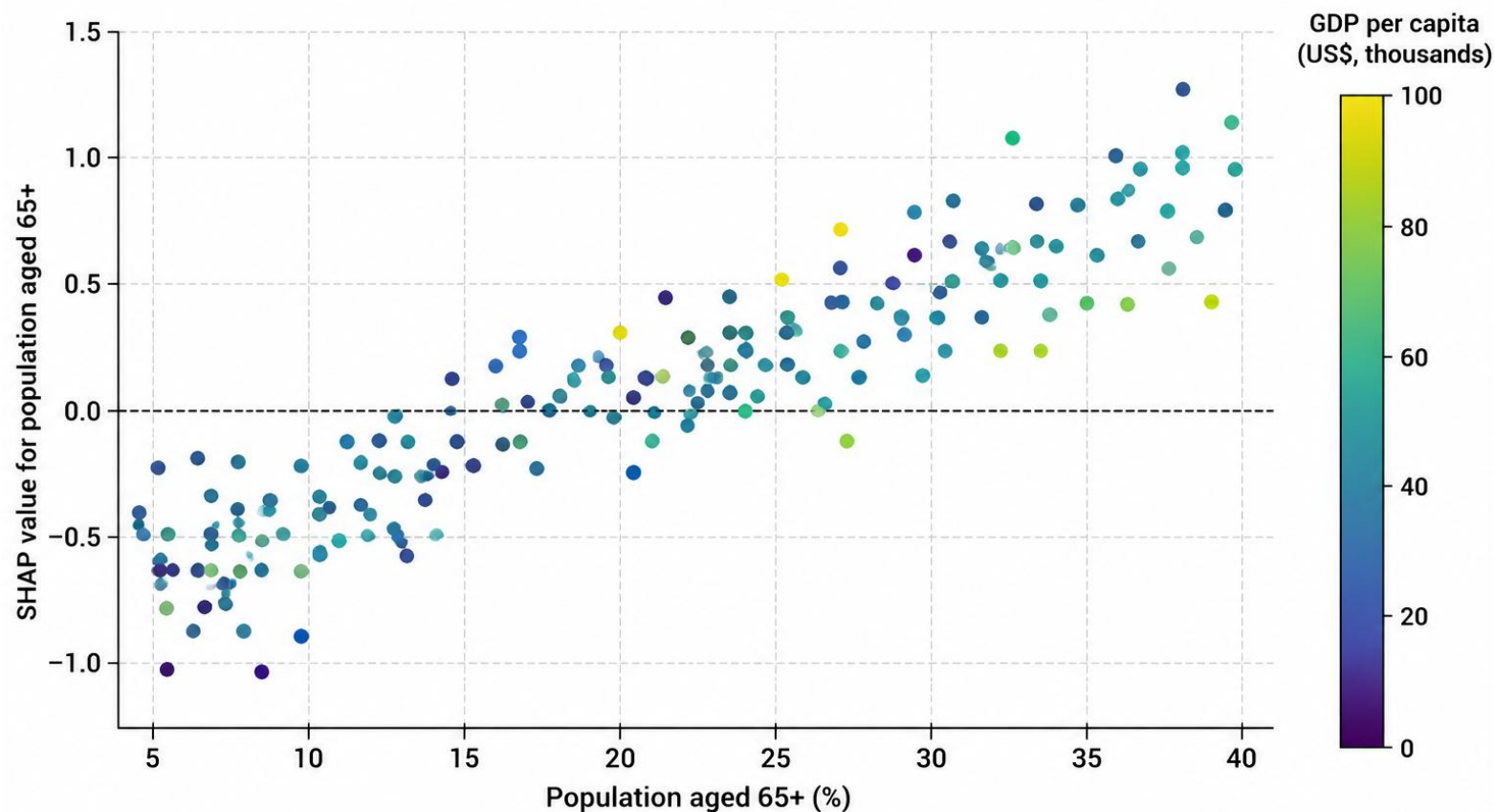
Model	Hyperparameters considered	Search domain	Planned/allowable trial budget	Optimization objective	Validation framework	Analytical role
Elastic Net	alpha; l1_ratio	Prespecified ranges corresponding to Table S14	30–50 trials	Minimize mean validation RMSE	Five expanding-window validation years, 2017–2021	Bayesian-optimization benchmark for the regularized linear model
Random Forest Regressor	n_estimators; max_depth; min_samples_split; min_samples_leaf; max_features	Prespecified ranges corresponding to Table S14	50–100 trials	Minimize mean validation RMSE	Five expanding-window validation years, 2017–2021	Bayesian exploration of nonlinear tree configurations
XGBoost Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_weight	Prespecified ranges corresponding to Table S14	50–100 trials	Minimize mean validation RMSE	Five expanding-window validation years, 2017–2021	Bayesian optimization of gradient-boosting configurations
LightGBM Regressor	learning_rate; max_depth; n_estimators; subsample; reg_lambda; min_child_samples	Prespecified ranges corresponding to Table S14	50–100 trials	Minimize mean validation RMSE	Five expanding-window validation years, 2017–2021	Bayesian optimization of leaf-wise boosting configurations
CatBoost Regressor	iterations; learning_rate; depth; l2_leaf_reg	Prespecified ranges corresponding to Table S14	50–100 trials	Minimize mean validation RMSE	Five expanding-window validation years, 2017–2021	Bayesian optimization of CatBoost complexity and regularization

Note: TPE = Tree-structured Parzen Estimator. This table documents the prespecified Optuna TPE optimization framework and allowable trial ranges considered during development-period tuning. Mean expanding-window validation RMSE was the intended optimization objective, and the 2022–2023 final test period was excluded from tuning. The reported trial ranges describe the tuning design and should not be interpreted as exact completed trial counts because complete trial-by-trial Optuna logs are not available in the current reproducibility materials. Accordingly, no convergence claim is made from this table.

Table S18. Prespecified pruning and early-stopping framework considered during development-period tuning

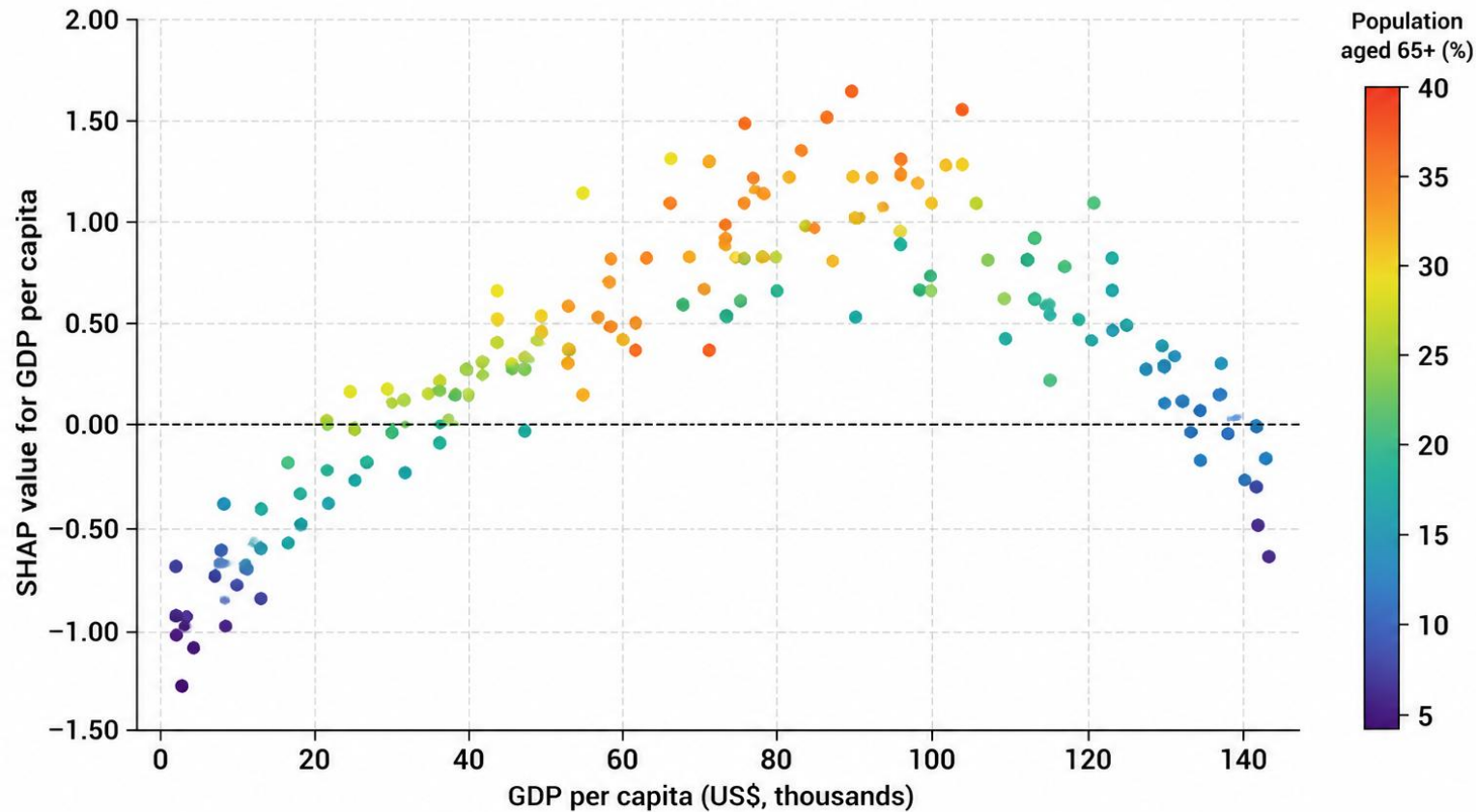
Model	Pruning/early-stopping approach considered	Applicability	Intermediate monitoring criterion	Validation framework	Intended purpose
XGBoost Regressor	Validation-based pruning or model-specific early stopping during boosting	High	Validation RMSE during successive boosting iterations	Expanding-window validation within the 2010–2021 development period	Identify weak candidate configurations early and reduce unnecessary computation
LightGBM Regressor	Validation-loss-based pruning or model-specific early stopping	High	Validation RMSE during successive boosting iterations	Expanding-window validation within the 2010–2021 development period	Improve computational efficiency while limiting unnecessary boosting iterations
CatBoost Regressor	Validation-based early stopping or trial-level pruning for weak configurations	Moderate to high	Validation RMSE during boosting progress and development-period validation	Expanding-window validation within the 2010–2021 development period	Reduce unnecessary computation while retaining promising candidate configurations
Random Forest Regressor	Trial-level screening after partial candidate evaluation where computationally feasible	Limited	Interim validation RMSE	Expanding-window validation within the 2010–2021 development period	Computational screening of weak configurations
Elastic Net	No substantive pruning required	Low	Validation RMSE	Expanding-window validation within the 2010–2021 development period	Pruning offers little practical benefit for the computationally inexpensive linear model

Note: This table documents the pruning and early-stopping framework considered for computational efficiency during development-period tuning. It does not represent a trial-by-trial execution log. Exact pruner type, stopping patience, warm-up steps, or model-specific pruning parameters should not be inferred where corresponding optimization logs are unavailable. No 2022–2023 observations were used for tuning, pruning, or early-stopping decisions. Pruning should therefore be interpreted as a development-stage computational strategy rather than an additional model-performance result.



Each point represents one of the 74 independent 2022–2023 test observations.
 Positive SHAP values increase the model prediction relative to its expected value,
 whereas negative values decrease it.

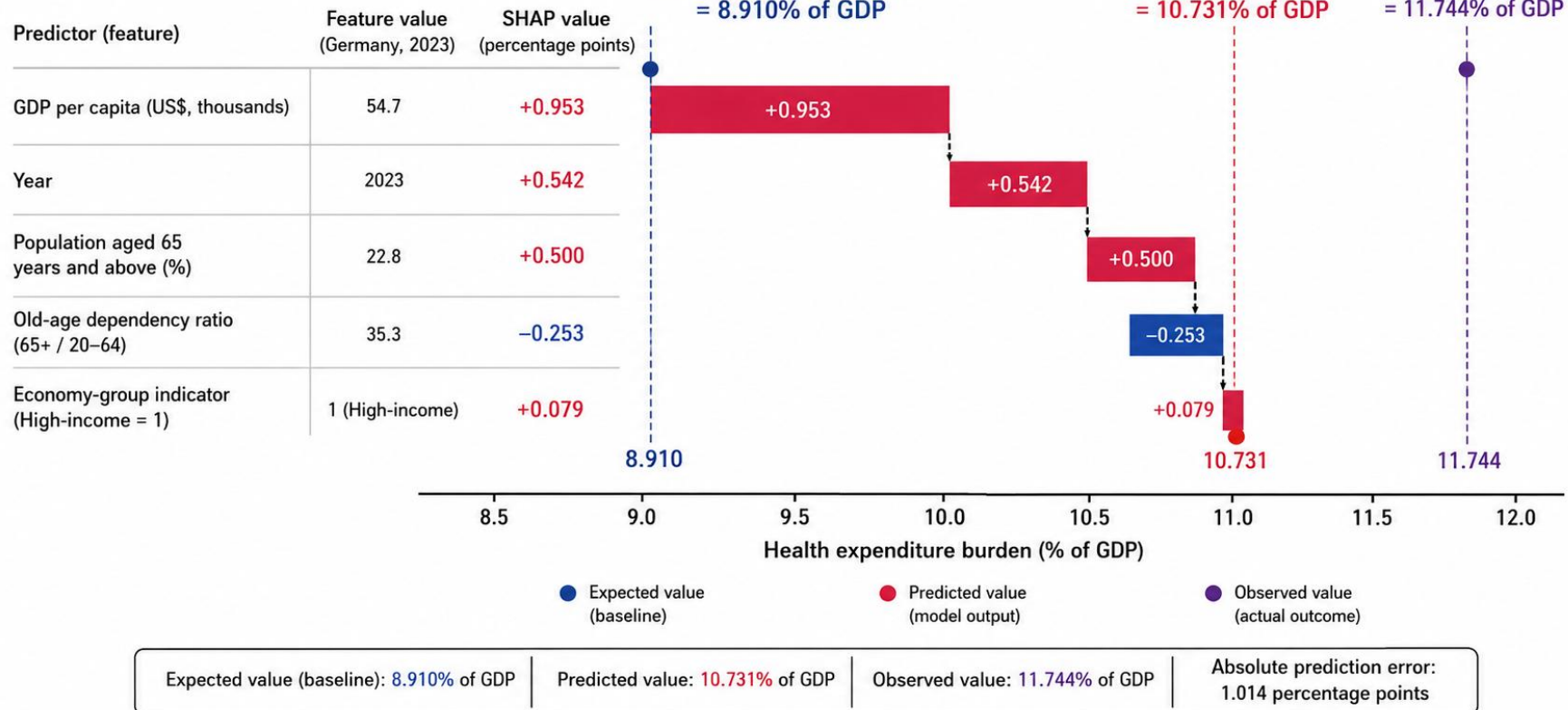
Figure S10. Detailed SHAP dependence of population aging in the validation-selected CatBoost model for the 2022–2023 independent test observations. Each point represents one independent test observation. The horizontal axis shows the population aged 65 years and above, the vertical axis shows its SHAP contribution to predicted health expenditure burden, and point color indicates GDP per capita. Positive SHAP values increase the model prediction relative to its expected value, whereas negative values decrease it. The pattern represents predictive model behavior and should not be interpreted as a causal effect.



Each point represents one of the 74 independent 2022–2023 test observations.
 Positive SHAP values increase the model prediction relative to its expected value,
 whereas negative values decrease it.

Figure S11. Detailed SHAP dependence of GDP per capita in the validation-selected CatBoost model for the 2022–2023 independent test observations. Each point represents one independent test observation. The horizontal axis shows GDP per capita, the vertical axis shows its SHAP contribution to predicted health expenditure burden, and point color indicates the population aged 65 years and above. Positive SHAP values increase the model prediction relative to its expected value, whereas negative values decrease it. The nonlinear pattern represents predictive model behavior and should not be interpreted as a causal income effect.

Local SHAP explanation for Germany, 2023



Notes: Positive SHAP values (red) increase the model prediction relative to the expected value.

Negative SHAP values (blue) decrease the model prediction.

Contributions are based on the final CatBoost model evaluated on the independent 2022–2023 test observations.

Figure S12. Local SHAP explanation for a representative independent-test observation: Germany, 2023. The observation was selected because its absolute prediction error was approximately the median among the 37 observations in the 2023 independent test year. The CatBoost model expected value was 8.910% of GDP and the predicted health expenditure burden was 10.731%, compared with an observed value of 11.744%. GDP per capita contributed +0.953 percentage points to the prediction, followed by year (+0.542), population aged 65 years and above (+0.500), old-age dependency ratio (–0.253), and the economy-group indicator (+0.079). Positive SHAP values increase the model prediction relative to its expected value, whereas negative values decrease it. The contributions describe local predictive model behavior and should not be interpreted as causal effects.

Table S19. Global SHAP importance of the validation-selected CatBoost model on the independent 2022–2023 test period

Rank	Predictor	Mean absolute SHAP	SD of absolute SHAP	Share of total mean absolute SHAP (%)	Interpretation
1	GDP per capita	0.843	0.470	36.1	Largest contribution to independent-test predictions; the dependence pattern is nonlinear across the observed income distribution.
2	Population aged 65 years and above	0.604	0.353	25.8	Second-largest predictive contribution; higher aging values generally contribute positively to predicted health expenditure burden.
3	Year	0.482	0.204	20.6	Material temporal contribution to predictions across the 2022–2023 independent test period.
4	Old-age dependency ratio	0.309	0.267	13.2	Moderate predictive contribution with a comparatively heterogeneous and non-monotonic dependence pattern.
5	Economy-group indicator	0.098	0.127	4.2	Small additional predictive contribution after incorporating demographic, macroeconomic, and temporal information.

Note: SHAP values were calculated for the 74 observations in the independent 2022–2023 test period after model-family and hyperparameter selection had been completed. The final CatBoost model was trained using the 2010–2021 development data with the selected configuration of 500 iterations, learning rate = 0.10, depth = 8, and L2 leaf regularization = 5. Mean absolute SHAP represents the average magnitude of each feature’s contribution to the model prediction, while the reported standard deviation describes variation in absolute SHAP magnitude across the independent test observations. SHAP values represent predictive contributions and should not be interpreted as causal effects, econometric coefficients, or measures of statistical significance. Government financing share and LTC workforce capacity are excluded because they are not predictors in the final core CatBoost model.

Table S20. Sequential conformal prediction performance for the validation-selected CatBoost model

Evaluation subset	N	Nominal coverage	Empirical coverage	Conformal half-width	Total interval width
2022	37	90%	86.5%	1.821	3.643
2023	37	90%	89.2%	1.897	3.794
Final-period overall	74	90%	87.8%	—	—
High-income economies	64	90%	85.9%	—	—
Emerging economies	10	90%	100.0%	—	—

Note: Absolute prediction residuals from expanding-window validation for 2017–2021 (N = 185) formed the initial conformity-score set. For the 2022 prediction set, the finite-sample conformal quantile was 1.821 percentage points, yielding a total interval width of 3.643 percentage points. The 2022 interval was generated without using 2022 outcomes. After the 2022 outcomes became available, the CatBoost model was refitted using observations through 2022, and the 37 realized 2022 absolute prediction errors were added to the calibration pool, increasing the conformity-score set to N = 222. The resulting 2023 conformal quantile was 1.897 percentage points, yielding a total interval width of 3.794 percentage points. Because calibration was updated sequentially, a single common interval width does not apply across the full 2022–2023 period or the pooled economy-group summaries. Coverage is therefore reported empirically for those pooled subsets. Temporal and cross-country dependence weaken the exchangeability assumptions

underlying standard conformal guarantees, so interpret the reported values as empirical calibration rather than exact finite-sample coverage guarantees. The emerging-economy estimate is based on only 10 independent-test observations and should therefore be interpreted cautiously.

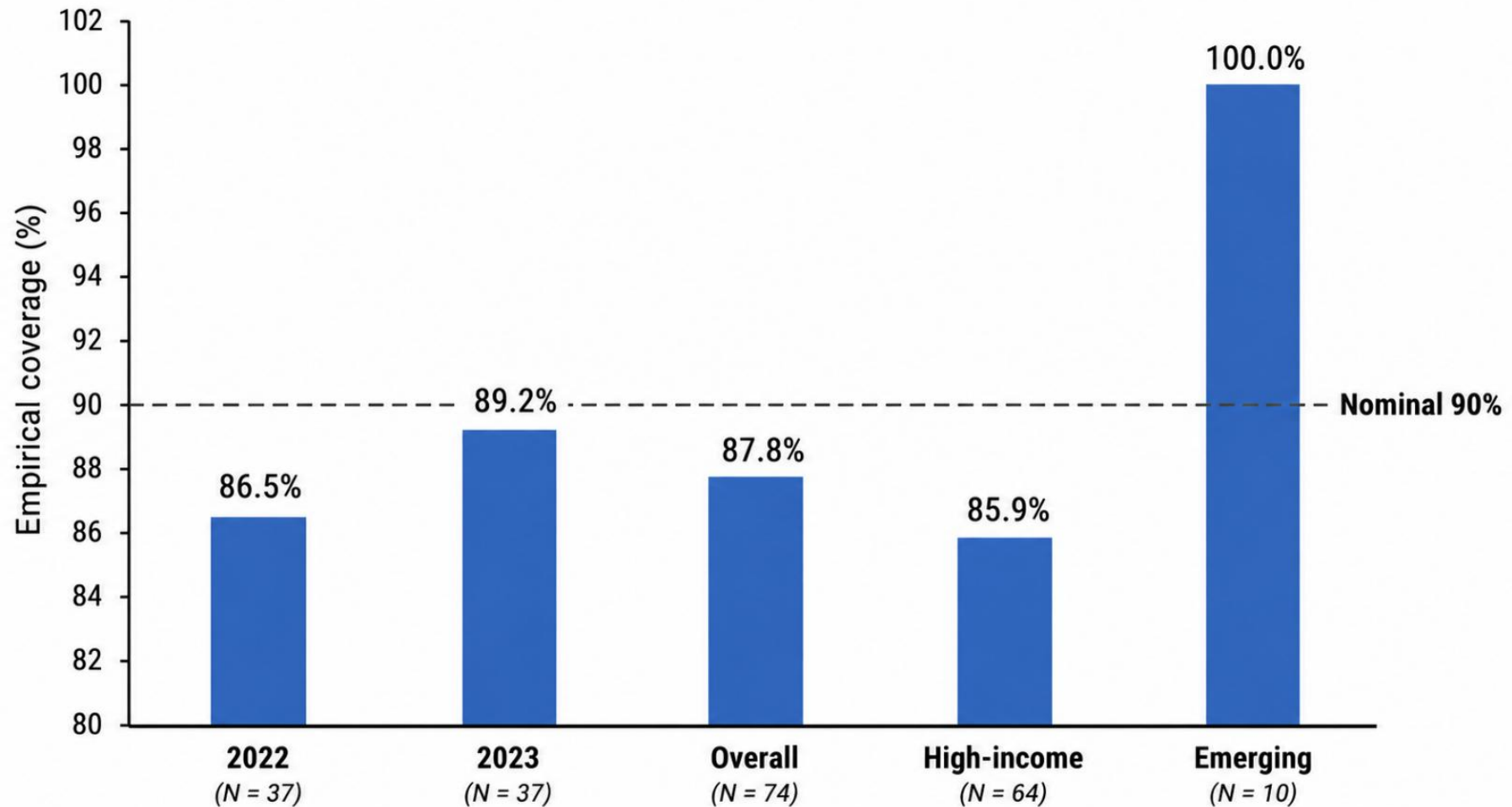


Figure S13. Empirical coverage of sequential 90% conformal prediction intervals. The dashed horizontal line represents the nominal 90% coverage level. Coverage is shown by evaluation year and economy group. The emerging-economy estimate is based on 10 independent-test observations and should be interpreted cautiously.

Table S21. Detailed assumptions for the 2024–2028 scenario sensitivity analysis.

Scenario	Demographic trend multiplier	GDP trend multiplier	Variables manipulated	Common starting point	Analytical role
Baseline	1.00	1.00	Pop65, Dependency, GDP_pc	Observed 2023 values	Reference pathway
Accelerated aging	1.50 for Pop65 and Dependency	1.00	Pop65, Dependency	Observed 2023 values	Demographic stress sensitivity
Moderated aging	0.75 for Pop65 and Dependency	1.00	Pop65, Dependency	Observed 2023 values	Lower demographic-growth sensitivity
Lower GDP growth	1.00	0.50	GDP_pc	Observed 2023 values	Macroeconomic downside sensitivity
Higher GDP growth	1.00	1.50	GDP_pc	Observed 2023 values	Macroeconomic upside sensitivity

Note: Country-specific historical log trends were estimated from the 2010–2023 observations. Government financing share and LTC workforce capacity are excluded from all core scenarios because they are not inputs to the scenario model and have insufficient full-panel coverage.

Table S21A. Rolling-origin backtesting of the anchored scenario projection model.

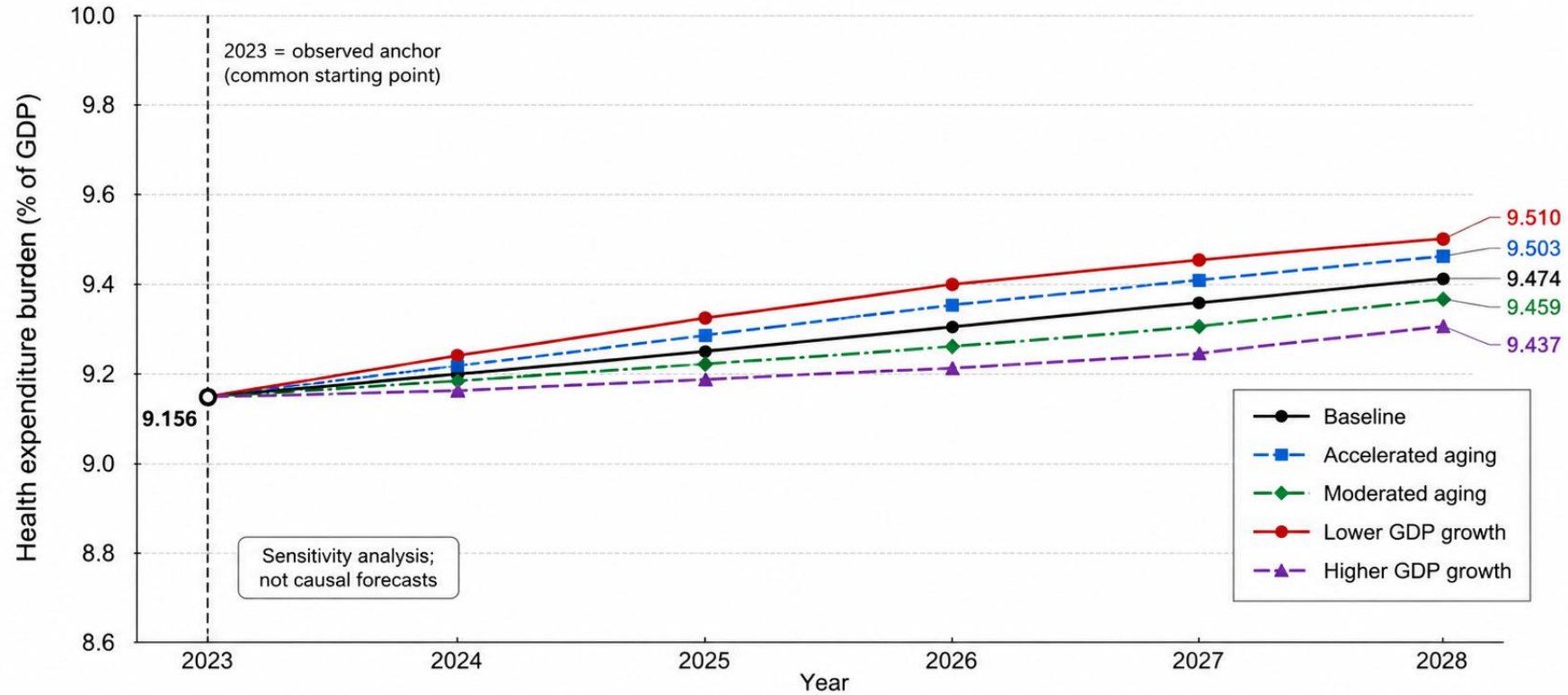
Forecast horizon	Number of historical origins	Scenario-model RMSE	Scenario-model MAE	Persistence RMSE	Persistence MAE
1 year	7	0.494	0.393	0.479	0.381
3 years	5	0.899	0.740	0.862	0.712
5 years	3	0.968	0.761	0.950	0.765

Note: Performance is comparable to, but does not systematically improve upon, persistence. Accordingly, the 2024–2028 outputs are interpreted as sensitivity trajectories rather than validated forecasts.

Table S21B. Observed predictor support and projected 2028 scenario ranges.

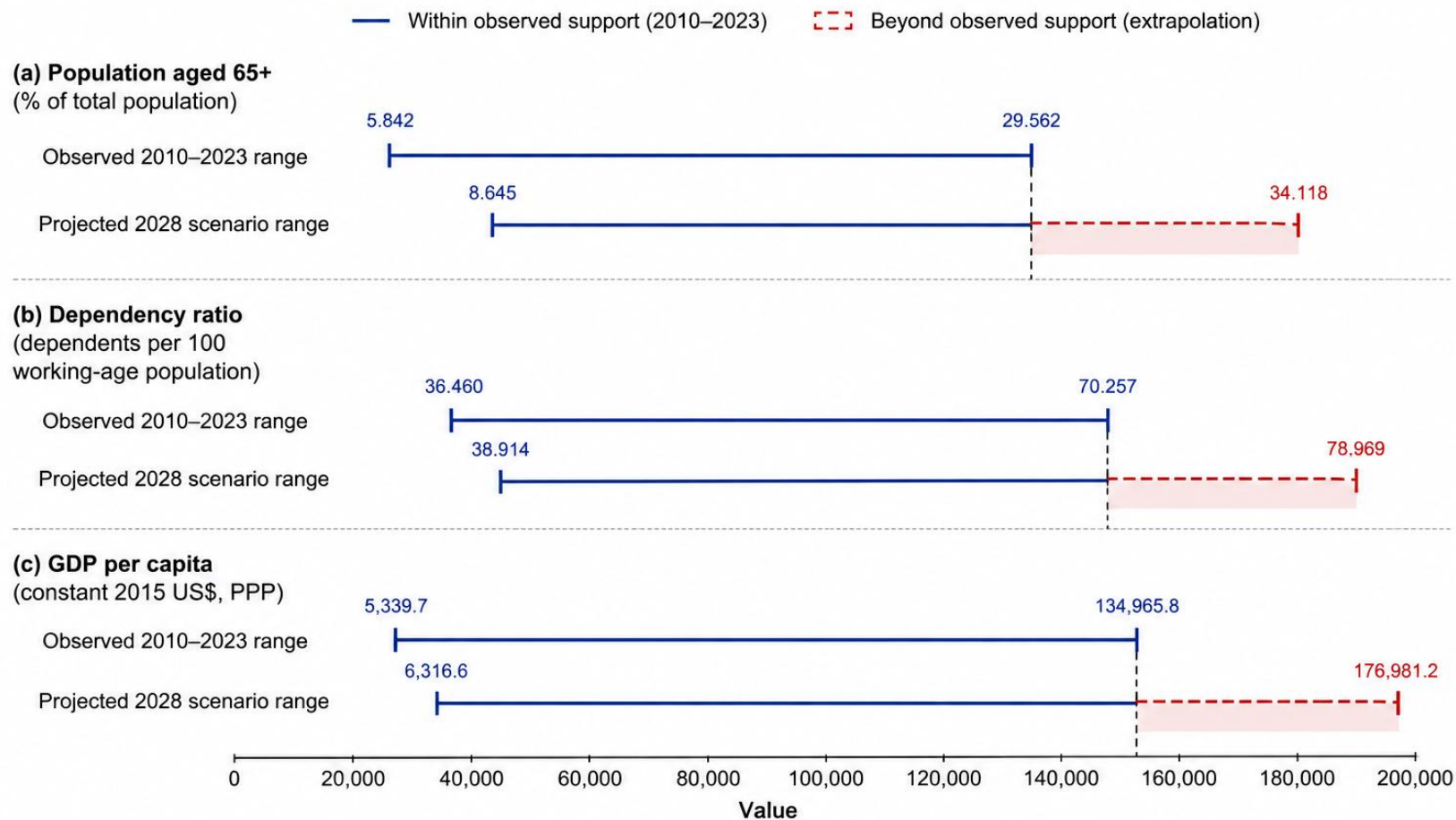
Predictor	Observed 2010–2023 minimum	Observed maximum	Projected 2028 minimum across scenarios	Projected 2028 maximum across scenarios
Population aged 65+	5.842	29.562	8.645	34.118
Dependency ratio	36.460	70.257	38.914	78.969
GDP per capita	5,339.7	134,965.8	6,316.6	176,981.2

Note: Some economy-specific projected predictor values exceed the observed training support even within the five-year horizon. This is explicitly considered when interpreting the scenario trajectories and is a principal reason the analysis is described as sensitivity analysis rather than unconditional forecasting.



Scenario	2023 (anchor)	2024	2025	2026	2027	2028
Baseline	9.156	9.225	9.293	9.358	9.417	9.474
Accelerated aging	9.156	9.236	9.316	9.392	9.453	9.503
Moderated aging	9.156	9.218	9.276	9.333	9.391	9.459
Lower GDP growth	9.156	9.246	9.335	9.418	9.474	9.510
Higher GDP growth	9.156	9.202	9.244	9.284	9.322	9.437

Figure S14. Detailed 2023–2028 scenario sensitivity trajectories for health expenditure burden. All scenarios are anchored to the observed 2023 sample mean and diverge only according to specified demographic and GDP-growth assumptions.



Note: Solid blue lines indicate the observed minimum and maximum values in the 2010–2023 panel. Projected 2028 scenario ranges are shown in blue where they fall within the observed support and in red shaded areas where they extend beyond the observed range. Values beyond the observed support indicate extrapolation and motivate interpretation of the scenario results as sensitivity analysis rather than unconditional forecasts.

Figure S15. Comparison of observed predictor support and projected 2028 scenario ranges. Projected values extending beyond the observed 2010–2023 range indicate extrapolation and support interpretation of the scenario outputs as sensitivity trajectories rather than unconditional forecasts.

Table S22. Summary of robustness and sensitivity analyses.

Robustness analysis	Sample/specification	Missing-data treatment	Main analytical purpose	Interpretation
R1 Core full-panel analysis	518 observations, 37 economies	No imputation required	Benchmark econometric and predictive analysis	Log-log estimates show weak and statistically imprecise within-country demographic effects; predictive relationships are evaluated separately
R2 Core-variable specification	HealthExp_GDP, Pop65, Dependency, GDP_pc	Complete variables retained directly	Confirms findings without sparse financing/workforce indicators	Core results are not dependent on Gov_Share or LTC_Workers
R3 Financing-augmented analysis	41 observations, 3 countries	Missing financing observations not imputed	Exploratory financing analysis	Insufficient coverage for general cross-country financing conclusions
R4 Workforce-augmented analysis	58 observations, 16 countries	Missing workforce observations not imputed	Exploratory workforce analysis	Provides supplementary service-capacity evidence only
R5 High-income subgroup	32 economies	Core complete variables	Descriptive/subgroup assessment	Higher observed expenditure burden and older demographic structure; no causal interpretation
R6 Emerging-economy subgroup	5 economies	Core complete variables	Exploratory subgroup assessment	Lower measured expenditure burden should not be interpreted as lower future care need or greater sustainability
R7 Derived-indicator sensitivity	Availability depends on parent variables	No synthetic full-panel completion	Descriptive sensitivity using FPI, affordability, public exposure, or workforce ratio	Derived indicators are deterministic transformations and are not independent evidence of fiscal sustainability
R8 Scenario sensitivity	2024–2028 anchored trajectories	Core predictors only	Evaluate demographic and GDP-growth sensitivity	Differences among pathways are modest; backtesting does not systematically outperform persistence; outputs are sensitivity trajectories rather than validated forecasts

Note: Robustness analyses distinguish full-panel evidence from restricted-sample supplementary evidence. No missing government-financing or LTC-workforce values are imputed to create full-panel predictors. Statistical association, predictive importance, and scenario sensitivity are interpreted as distinct forms of evidence.

Table S23. Independent final test evaluation of the validation-selected CatBoost model.

Model	Development period	Selection basis	Final test period	Test N	RMSE	MAE	sMAPE (%)	R²
CatBoost	2010–2021	Lowest expanding-window validation RMSE	2022–2023	74	1.692	1.187	13.022	0.470

Note: No candidate-model ranking was performed using the final test observations. Model-family and hyperparameter selection were completed using only expanding-window validation within the development period.

Table S24. Reproducibility files and repository contents

Repository component	File/content type	Purpose
Cleaned analytical dataset	CSV/Excel dataset	Provides the harmonized country-year data used for descriptive statistics, group comparisons, econometric analysis, machine-learning inputs, SHAP analysis, conformal evaluation, and scenario sensitivity.
Data dictionary	CSV/Excel table	Defines study variables, source-variable meanings, units, analytical roles, and interpretation limits.
Variable-construction documentation	Python script and supporting documentation	Documents construction of HealthExp_GDP, Public_Exposure, FPI, Worker_Ratio, and Affordability from the harmonized source variables.
Missing-data diagnostics	Tables and figures	Documents variable-level and country-year completeness, including restricted coverage of government-financing and LTC-workforce variables.
Econometric analysis files	Scripts, tables, and supporting documentation	Supports reproduction of pooled, fixed-effects, and random-effects log-log models, within-between elasticity decomposition, and associated model diagnostics.
Machine-learning analysis files	Python scripts and output tables	Supports reproduction of expanding-window validation, candidate-model comparison, final CatBoost fitting, and independent 2022–2023 test evaluation.
Hyperparameter-tuning documentation	Search-space tables and available tuning records	Documents the prespecified Grid Search, Random Search, Optuna TPE, and pruning/early-stopping frameworks used or considered within the 2010–2021 development period. Report exact trial-by-trial records only where available.
SHAP explainability files	SHAP scripts, tables, figures, and underlying test-observation data	Supports reproduction of global SHAP importance, dependence plots, interaction analysis, and local explanation for the independent 2022–2023 test observations.
Conformal-prediction files	Python scripts and output tables	Supports reproduction of the sequential 90% conformal procedure, including 2022 calibration, model refitting through 2022, 2023 calibration updating, interval widths, and empirical coverage.
Scenario-sensitivity files	Scripts, tables, and figures	Supports reproduction of the anchored 2024–2028 demographic and GDP-growth sensitivity trajectories, rolling-origin backtesting, and predictor-support diagnostics.
Supplementary output files	Word/Excel/PDF/figure files	Provides supporting tables, figures, robustness summaries, diagnostics, and methodological documentation corresponding to the Supplementary Materials.
Repository documentation	README and reproducibility notes	Describes the study scope, analytical workflow, file structure, interpretation limits, and steps required to reproduce the available analyses.

Note: The public repository provides cleaned analytical data, a documented analytical workflow, available scripts, output tables, and supporting reproducibility materials associated with the study. The primary outcome is health expenditure burden, not direct LTC expenditure or fiscal sustainability. Government-financing and LTC-workforce variables remain restricted-coverage supplementary indicators and are not imputed to construct full-panel predictors. Hyperparameter-search tables document the prespecified development-period tuning frameworks; do not infer exact trial-by-trial tuning histories where corresponding logs are unavailable. Interpret repository outputs as authoritative only when they are generated from the documented analytical data and code and are consistent with the manuscript and Supplementary Materials.